

Project 3

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Branch :Computer Engineering

Domain:AI & ML Development

Code:

```
"""Project 3: Personal Expense Tracker """
print("!#####**<&>***#####!")
print("Personal Expenct tracker:")
print("!#####**<&>***#####!")

class Trace():
    def __init__(self,budget):
        self.budget = budget
        while budget <0:
            self.budget = float(input("Budget should be positive number:"))

    def addexpenct(self,dit):
        amount = int(input("Enter the amount of expence:"))
        while amount <0:
            amount = float(input("Budget should be positive number:"))
        category = input("Enter the category:")
        date =input("Enter the date of expence:")
        discription = input("Enter the discription of expence:")
        if category not in dit:
            dit[category] ={}
        if date not in dit[category].keys():
            dit[category][date] = {}
        dit[category][date][amount]={"amount":amount,"discription":discription}

    def view_expences(self,dit):
        for i in dit.keys():
            print(f"Category:{i}---->")
            for j in dit[i].keys():
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for k in dit[i][j].keys():
    print("***30)
    print(f" Date      :{j}")
    print(f" Amount    :{dit[i][j][k]['amount']}")
    print(f" Discprition:{dit[i][j][k]['discription']}")
    print("***30)

```

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def budget_report(self,dit):
    total = 0
    for i in dit.keys():
        print(f"Category:{i}---->")
        for j in dit[i].keys():
            for k in dit[i][j].keys():
                print("***30)
                print(f" Date      :{j}")
                print(f"Amount:{dit[i][j][k]['amount']}")
                total += dit[i][j][k]['amount']

    report = self.budget - total
    if report >0:
        print("Good Budget!")
    else:
        print("You need more planing for spending!")

```

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def search(self,dit,category):
    if category in dit:
        print("Expences are as follows:")
        for j in dit[category].keys():
            for k in dit[category][j].keys():
                print("***30)
                print(f" Date      :{j}")
                print(f" Amount    :{dit[category][j][k]['amount']}")
                print(f" Discprition:{dit[category][j][k]['discription']}")
    else:
        print("Category is not present in the Expence list:")

```

```

def topcategory(self,dit):
    cat = []
    cat.extend(dit.keys())
    amt = []
    srt = dict()

```

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for i in dit.keys():
    for j in dit[i].keys():
        amt.extend(dit[i][j].keys())
    amt.sort()
    srt[i]=amt[-1]

cat.clear()
amt.clear()
amt.extend(srt.values())
amt.sort()
for i in srt.keys():
    if srt[i] == amt[-1]:
        print(f"Top Category is:{i}")
        print(f"Expendature is:{amt[-1]}")

```

```

def main():
    t1 = Trace(int(input("Enter the budget:")))
    exp = dict()
    choice = 0
    print("1.add Expence          2.View Expence          3.Budget report          4.Search")
    print("5.Top category          -1.Exit")
    while True:
        choice = int(input("Enter the choice:"))
        print("Enter the choice:")
        if choice==1:
            t1.addexpence(exp)
        elif choice ==2:
            t1.view_expences(exp)
        elif choice ==-1:
            break
        elif choice == 3:
            t1.budget_report(exp)
        elif choice ==4:
            t1.search(exp,input("Enter the category:"))
        elif choice == 5:
            t1.topcategory(exp)
        else:
            print("Invalid Choice:")
    print("Loop terminated!")

```

main()

Output:

```

PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Local\Python\pythoncore-3.14-64\python.exe "d:/python/ct3.py"
#####<&>#####!
Personal Expençe tracker:
#####<&>#####!
Enter the budget:100000000
1.add Expençe          2.View Expençe          3.Budget report          4.Search
5.Top category        -1.Exit
Enter the choice:1
Enter the choice:
Enter the amount of expençe:10000
Enter the category:child fund
Enter the date of expençe:12/5/2028
Enter the discription of expençe:primary fund for child with no parent
Enter the choice:1
Enter the choice:
Enter the amount of expençe:1000
Enter the category:food
Enter the date of expençe:12/7/2025
Enter the discription of expençe:lunch due sunday and mess tiffin
Enter the choice:1
Enter the choice:
Enter the amount of expençe:100000
Enter the category:hospital
Enter the date of expençe:5/5/2027
Enter the discription of expençe:free treatment for people suffering from serious disease's
Enter the choice:2
Enter the choice:
Category:child fund--->
*****
Date      :12/5/2028
Amount    :10000

```

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```
PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Local\Python\pythoncom
ct3.py"
```

```
Amount      :10000
```

```
Discription:primary fund for child with no parent
```

```
*****
```

```
Category:food---->
```

```
*****
```

```
Date        :12/7/2025
```

```
Amount      :1000
```

```
Discription:lunch due sunday and mess tiffin
```

```
*****
```

```
Category:hospital---->
```

```
*****
```

```
Date        :5/5/2027
```

```
Amount      :100000
```

```
Discription:free treatment for people suffering from serious disease's
```

```
*****
```

```
Enter the choice:3
```

```
Enter the choice:
```

```
Category:child fund---->
```

```
*****
```

```
Date        :12/5/2028
```

```
Amount:10000
```

```
Category:food---->
```

```
*****
```

```
Date        :12/7/2025
```

```
Amount:1000
```

```
Category:hospital---->
```

```
*****
```

```
Date        :5/5/2027
```

```
Amount:100000
```

```
Good Budget!
```

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```
PS D:\python programs\Industrial traning> & C:\Users\prita\AppData\Local\Python\pythoncore
ct3.py"
```

```
Enter the choice:
```

```
Category:child fund---->
```

```
*****
```

```
    Date      :12/5/2028
```

```
Amount:10000
```

```
Category:food---->
```

```
*****
```

```
    Date      :12/7/2025
```

```
Amount:1000
```

```
Category:hospital---->
```

```
*****
```

```
    Date      :5/5/2027
```

```
Amount:100000
```

```
Good Budget!
```

```
Enter the choice:4
```

```
Enter the choice:
```

```
Enter the category:child fund
```

```
Expences are as follows:
```

```
*****
```

```
    Date      :12/5/2028
```

```
    Amount    :10000
```

```
    Discription:primary fund for child with no parent
```

```
Enter the choice:5
```

```
Enter the choice:
```

```
Top Category is:hospital
```

```
Expendature is:100000
```

```
Enter the choice:-1
```

```
Enter the choice:
```

```
Loop terminated!
```

```
PS D:\python programs\Industrial traning>
```